Cherry Bhatt

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EDUCATION

Carnegie Mellon University (2023-2025) Pittsburgh, USA
Master of Science in Robotic Systems Development, School of Computer Science
Learning for 3D Vision | Visual Learning and Recognition | Computer Vision | Robot Autonomy | Robot Mobility | Systems Engineering | | Manipulation, Estimation and Control

University of Mumbai (2019-2023) Mumbai, India
Bachelor of Technology in Electronics Engineering, DJ Sanghvi College of Engineering
Embedded Systems and RTOS | Control Systems and Instrumentation | Digital Image and Signal Processing | AI and ML | Robotics and Automation

PROJECTS

Learning for 3D Vision, Carnegie Mellon University January 2025 - May 2025
- Reduced model latency by 50% for BEV map generation for roadside 3D scenes leveraging TensorRT
- Model quantization by altering precision modes and batch sizes for edge deployment

Monocular 3D Detection, Carnegie Mellon University January 2024 - May 2024
- Developed YOLOv7-based architecture for monocular 3D bounding box detection in autonomous driving systems
- Achieved 92.1% mAP for cars, 68.3% for pedestrians, and 61.0% for cyclists on KITTI dataset

Autonomous Reforestation Robot, Carnegie Mellon University August 2023 - December 2024
- Built complete perception stack on UGV, enabling 10 plantings/20 minutes through environmental understanding
- Implemented sensor fusion pipeline integrating ZED cameras and IMU data for 3D obstacle detection

Decluttering Robot Arm, Carnegie Mellon University January 2024 - May 2024
- Deployed collision-free trajectory planning on Franka robot arm for autonomous object manipulation tasks
- Implemented vision-guided grasping pipeline using object segmentation and depth-based 3D pose estimation

EXPERIENCE

Vision and Learning Researcher, Kantor Lab (CMU) May 2024 - August 2024
- Designed a PyTorch-based ML pipeline for FireBlight detection preventing crop loss in apple orchards, achieving 85% accuracy on raw field images
- Built automated Tree of Heaven bounds identification system using VLMs to support lanternfly population control initiatives in Pittsburgh region

Autonomy Software Team Member, Moonranger (CMU X NASA) August 2024 - December 2024
- Identified critical system vulnerabilities in lunar rover operations and developed a 13-case contingency plan tackling unexpected communication failures
- Implemented resilient core Flight System software ensuring power failure recovery, to meet mission-critical standards

IoT Research Fellow, TIH (IIT Bombay) January 2022 - December 2022
- Secured $7,000 research grant to develop EMG-controlled wheelchair, implementing real-time signal processing and sensor integration on ESP32 with multithreading optimization
- Delivered functional prototype achieving 60% signal detection accuracy and 3-hour battery life through embedded programming and power management

Computer Vision Intern, Lifespark Technologies January 2023 - February 2023
- Deployed and fine-tuned 3D pose detection pipeline using OpenPose3D and MMpose frameworks for gait analysis
- Optimized keypoint detection accuracy for pre-clinical assessment of Parkinson's disease

SKILLS

Frameworks: ROS2 | Git | PyTorch | Tensorflow | TensorRT | core Flight System | RTOS
Programming Languages: Python | C/C++ | MATLAB | Bash
Platforms: Unity3D | AWS | Docker | Mujoco | Linux | EagleCAD | NVIDIA Isaac SDK